

AGORA

One Sentence, One Living World

Generating Persistent Worlds for Human–AI Social Interaction

Boxun Xu Yueze Wang Peng Li

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Abstract. Generating a social world requires inhabitants to change shared spatial, material, and social state. We present Agora, a system that compiles a short premise into a persistent rendered environment with roles, institutions, possessions, and private knowledge. Humans and language agents use generated action opportunities or propose context-specific actions, whose effects a World Coordinator adjudicates against shared state. A masked model-comparison survey yields ten eligible returns, forty image comparisons, and judgments on all 21 pairs among seven language-model configurations using a shared pipeline and premise. Preferences differ by criterion: Gemini 2.5 Flash receives 8/10 exploration preferences, while GPT-5.6 Terra and Sol receive 11/13 and 9/11 map-readability preferences, respectively. These sparse fixed-artifact judgments do not establish a general model ranking. An internal generation control records higher completion for the staged procedure at higher cost; its structure judgments are reported separately and do not benchmark Agora against independent systems. Five two-round interaction runs produce 53 events and 49/56 successful action results. In a retrospective audit of 21 scripted 24-round trajectories, 1,365/1,512 actions execute successfully, but only 398 successes change inventory or location. Together, these findings distinguish construction reliability, artifact preference, and enacted consequences. We release inspectable world artifacts, returned survey analysis, and reproducible evidence; behavioral fidelity and useful human collaboration remain unestablished.



Figure 1. Five generated settings share a world schema while varying institutions, spaces, and visual identity. Panels show preserved map assets and representative character frames; they are not new experimental outputs.

1 Introduction

A generated social world needs persistent consequences. In a floating embassy where endangered languages are political assets, inhabitants must be able to access archives, trade translation rights, and respond when a flood threatens the only surviving treaty. These activities require explicit places, possessions, information, and rules behind the rendered scene.

Several research lines address parts of this problem. Procedural content generation studies representations and criteria for functional game content [1, 2]. Word2World and DreamGarden extend language-based authoring to playable games and development plans [3, 4]. Generative Agents demonstrates how local memory and encounters can produce community activities in an authored town [5], while Concordia grounds language-described actions through a Game Master [6]. These advances motivate a joint design question: how can one short premise specify both a world and the conditions under which a society can act within it?

We present Agora, a system that compiles a one-sentence premise into a persistent, visually grounded environment for humans and language agents. Its representation connects rooms, roles, institutions, possessions, private knowledge, vi-

sual assets, and action opportunities. A generated repertoire makes the world’s activities discoverable; an open-proposal channel lets an actor express a more specific intention. A World Coordinator interprets each proposal against shared constraints before its effects enter the world state. For example, proposing a treaty exchange does not by itself transfer ownership: the resources, participants, and required acceptance must support the change.

The research object is a state trajectory

$$\tau = (S_0, q_1, d_1, S_1, \dots, q_T, d_T, S_T), \quad (1)$$

where q_t is an intended action and d_t is the coordinator decision. This representation distinguishes an intention, an enacted effect, and a later response. It also makes explicit which observations concern software execution, simulated social behavior, or human experience.

Our contributions are threefold:

1. A compositional language-to-world pipeline connecting semantic, visual, spatial, and mutable social state, with dependency-directed local refinement.
2. A shared action interface for humans and agents that combines generated opportunities with coordinator-governed

proposals and persistent consequences.

3. Audited evidence from ten paired world briefs, a separate 24-cell one-sentence generation corpus, five locally observed interaction runs, and 21 scripted execution trajectories, plus ten eligible visual-survey returns covering image and structure preferences. The results distinguish construction reliability, human artifact preference, and enacted consequences.

The completed studies characterize an authoring and experimentation platform. A paired memory-intervention design remains prospective (Appendix E); behavioral fidelity and causal social effects require further evidence.

2 Related Work and Research Position

2.1 From Generated Content to Executable Worlds

Procedural content generation distinguishes what is represented, how it is generated, and how its fitness is evaluated [1]. PCGML further connects learned content models with repair and critique [2]. This lineage matters for Agora: decomposing a world and repairing a faulty part are established design ideas. Our use of them ties a repair to explicit dependencies among role, inventory, geometry, media, and runtime state.

Word2World converts stories into playable tile-based games and evaluates its generation stages [3]. DreamGarden recursively decomposes a user prompt into plans executed by specialized modules, with user feedback during Unreal Engine development [4]. Both are close precedents for prompt-driven, staged authoring. Agora emphasizes a reusable social-state representation: the generated institutions and objects become inputs to ownership, knowledge access, and action adjudication during play. We make no claim that prompt-based world generation or hierarchical planning is new.

Environment generation also serves embodied-agent training. ProcTHOR samples diverse interactive houses [7]; Holodeck uses language and spatial constraints to produce 3D scenes from prompts [8]. Genie instead learns action-controllable visual dynamics from video [9]. These approaches provide distinct substrates for interaction. Agora uses explicit entity identities and state transitions in a rendered 2D environment, making social and material changes directly inspectable. Visual generation and symbolic persistence address complementary aspects of world construction.

2.2 Grounded Dialogue and Shared Play

TextWorld provides generated text games with explicit state tracking and controllable task difficulty [10]. ScienceWorld evaluates whether agents can carry out scientific procedures in an interactive environment [11]. Both motivate evaluating executed consequences rather than the plausibility of a textual plan.

LIGHT is a particularly relevant predecessor: humans and models inhabit a fantasy environment, combining dialogue with actions grounded in local objects and characters [12]. Thus shared human-agent play and situated dialogue are not unique to Agora. Our focus is generating the surrounding social and material setting from a premise, then coupling visible activities and novel proposals to the same persistent state. Human-

directed events in our logs establish reachability of the player, while experience and collaboration quality require a participant study.

2.3 Persistent Communities and Social Simulation

Generative Agents combines observations, retrieval, reflection, and planning in a town of 25 agents, illustrating how remembered encounters can support coordinated activities [5]. Concordia provides a flexible simulation framework in which agents describe intentions and a Game Master grounds their effects [6]. The World Coordinator follows this separation of intention and environment response; Agora connects it to premise-generated identities, inventories, rules, and rendered spaces.

AgentSociety studies large populations and interventions in a societal environment [14]; LMAgent combines multi-modal shopping behavior with large-scale social interaction [15]. Avalon studies cooperation and confrontation under explicit game rules [16]. These systems show why trajectories and institutions matter beyond isolated dialogue quality. Our smaller studies address authoring diversity and trace inspectability, without claiming comparable scale or empirical correspondence to society.

2.4 Action, Feedback, and Feasibility

ReAct interleaves reasoning and environment actions [17]; Reflexion stores verbal feedback for subsequent attempts [18]; Voyager accumulates executable skills through environment feedback and verification [19]. These methods motivate the observation-intention-outcome cycle. SayCan explicitly connects language-level usefulness to skill feasibility [20]. Agora applies the analogous distinction to social and material effects: an intention can be intelligible while its requested transfer, construction, or status change is infeasible. Its action interface adds partner acceptance, ownership, and world-specific rules to that decision. This is a system design connection, not a claim of algorithmic equivalence or an empirical comparison with those agents.

2.5 Evaluation and Validity

SOTOPIA evaluates interaction through multiple social goals and criteria [13]. Subsequent work shows that an omniscient simulation can appear more successful than an information-asymmetric interaction [21]. This motivates our explicit observation boundary and private knowledge, but does not by itself validate their implementation. SOTOPIA- π also reports evaluator overestimation of agents trained for social interaction [22]. Together with broader LLM-judge biases [23], this motivates keeping structural checks, state-derived metrics, and the present human artifact judgments separate.

AgentBoard motivates inspecting intermediate progress rather than relying on final success alone [24]. We similarly report executed changes separately from plausible intentions, but do not claim that our mutation counts measure validated task progress. AI Agents That Matter emphasizes cost, hold-outs, and reproducibility [25]; this informs our separation of unequal-budget architecture comparisons from model coverage audits and our explicit treatment of retrospective evidence.

The ODD protocol emphasizes explicit entities, schedul-

Research line	Representative substrate	Action or generation mechanism	Relation to Agora
Word2World; DreamGarden [3, 4]	Generated game content and environments	Story expansion; hierarchical development plans	Precedents for language-driven, staged authoring
ProcTHOR; Holodeck [7, 8]	Procedural or language-conditioned 3D scenes	Interactive objects and spatial constraints	Physical environment diversity and grounding
Genie [9]	Learned visual dynamics	Latent, frame-level control	Complementary representation of world dynamics
LIGHT; SOTOPIA [12, 13]	Grounded fantasy play; social scenarios	Dialogue, actions, and social goals	Shared interaction and social evaluation
Generative Agents; Concordia [5, 6]	Persistent communities; grounded simulations	Memory and planning; Game Master adjudication	Direct precedents for memory and action grounding
AgentSociety; LMAgent [14, 15]	Societal and e-commerce environments	Large populations and domain-specific interactions	Scale and behavioral evaluation beyond this study
Agora	Premise-generated 2D social worlds	Typed shared state; repertoire and open proposals	Connects world authoring to inspectable social trajectories

Table 1. Research positioning by representation and mechanism. Entries describe the cited systems’ emphasis, not an exhaustive feature audit or a performance ranking. We do not treat unreported capabilities as absent.

ing, initialization, and model rationale for reproducible agent-based modeling [26]. We adopt these reporting priorities in Appendix A. Human-grounded agent simulation, such as models built from interviews and surveys and evaluated against the people they represent [27], sets a different evidentiary standard from fictional world coherence. Finally, work on complex contagion distinguishes information transmission from socially reinforced adoption [28]. Accordingly, the prospective intervention measures both warning diffusion and material treaty rescue; neither network density nor fluent conversation substitutes for coordinated action.

3 The Agora System

3.1 World Representation

Given a one-sentence premise p , the generation pipeline constructs

$$W = (E, G, R, K, A, M, P), \quad (2)$$

where E contains entities and their stable identities; G is traversable geometry; R contains physical, economic, and institutional rules; K specifies knowledge allocation and access; A is the initial action repertoire; M associates visual media with entities; and P records world identity and revision. Mutable state S_t contains locations, holdings, knowledge acquired during play, tasks, relationships, and event status. The schema is shared across worlds; its entities and rules depend on p .

This separation gives language-generated descriptions operational meaning. An archive must refer to an existing room, an item must have a valid owner, and a private fact must have a defined holder. Each visual asset is linked to the same entity identity used by the runtime. Changes then propagate through the state observed by subsequent actors rather than remaining only in dialogue.

3.2 Compositional Generation and Local Refinement

A planning stage identifies institutions, conflicts, resources, and activity loops. Specialized stages expand materials, rooms, items, roles, main characters, knowledge, visual direction, wardrobe, property, and placement. The shared representation constrains references between their outputs. Visual generation

produces room floors, character atlases, and objects under a common style specification; assembly connects these assets to the playable geometry and entity identities.

Validation targets four properties: semantic completeness, cross-modal identity, navigable and readable space, and persistent interactive state. Structural validators check references, inventories, asset provenance, and image contracts. Visual inspection and interactive checks supply different evidence about readability and use. When a check identifies an invalid character–item relation, that dependency can be revisited while unaffected stages remain fixed. This instantiates repair as a constrained part of generation, consistent with the broader PCG literature [2].

Running example. Tidal Embassy of Lost Languages turns linguistic scarcity into institutions and actions. Its generated configuration contains eight rooms, 25 named main characters, and 20 item types. Diplomats, smugglers, and memory divers receive different possessions and private knowledge. The setting exposes activities such as translating an archaic bribe, forging a linguistic seal, and rescuing a flooded treaty. These relations create reasons to move, exchange, withhold information, and seek cooperation. Appendix G follows the full generation trace, including failed room images and localized visual repair.

3.3 Bounded Observation and Situated Intention

At round t , an activation policy selects agents. Agent i receives

$$o_i^t = (\ell_i^t, N_i^t, I_i^t, K_i^t, H_i^t, Q_i^t), \quad (3)$$

comprising location, nearby actors, inventory, accessible knowledge, episode and task memory, and selected directed relationships. Role and goals provide the decision context. The observation boundary is intended to preserve information asymmetry, an important distinction from omniscient social simulation [21].

The agent chooses a partner and an intention, using either a generated action or an open proposal. Visible actions explain what an institution makes possible; proposals express intentions absent from that initial vocabulary. Open proposals in the audited implementation support four effect primitives:

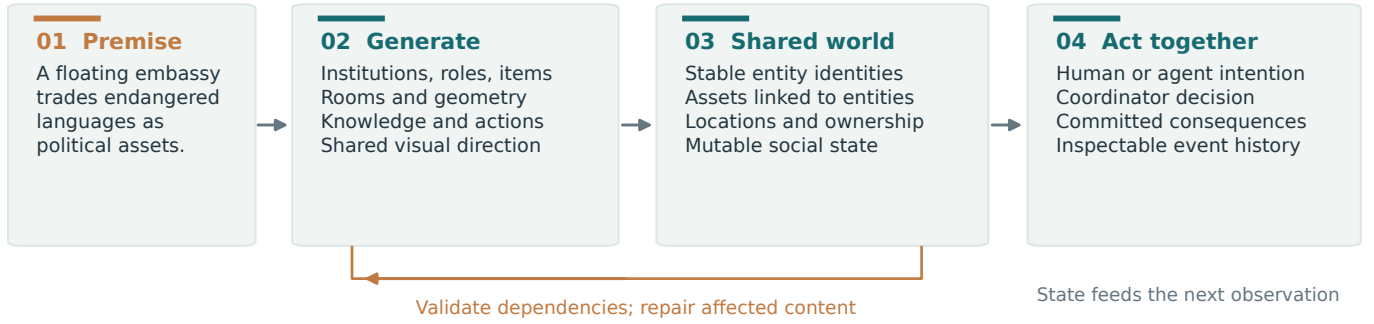


Figure 2. From premise to persistent consequences. Specialized generation stages populate a shared world representation. Validation connects semantic, visual, and spatial requirements; local repair revisits affected dependencies. Humans and agents act through the same world, while logged decisions and state changes support separate construction and interaction evaluations.

Effect	Required grounding
Speech	Addressable participant and a permitted channel
Transfer	Existing item, ownership, quantity, and co-presence where required
Trade	Transfer conditions and the partner’s acceptance
Creation	Declared available inputs and a resulting object identity
Status	Permitted target, bounded magnitude, and duration
Movement	Reachable destination and valid occupancy

Table 2. Examples of action conditions in the shared world. Linguistic plausibility alone does not establish that a requested effect is executable.

speech, transfer, item creation, and temporary status. Movement is a separate action route. These primitives can describe new contextual intentions, but they are not arbitrary executable programs or a general physics engine. Humans can approach an inhabitant, send messages, use or trade items, select a visible action, or formulate a proposal through the same world interface. This extends the shared grounded-play tradition of LIGHT [12] to premise-generated settings.

3.4 Coordinator-Governed State Changes

We write an open proposal as $q = (i, j, u, F, C)$: actor i , optional target j , language intention u , requested effects F , and declared consent evidence C . The model supplies a structured proposal; the audited coordinator core uses deterministic schema and state checks to return a decision and explanation,

$$\Gamma(W, S_t, q) = (d, r), \quad d \in \{\text{approve, revise, reject}\}. \quad (4)$$

Following the action-grounding role of Concordia’s Game Master [6], interpretation is separated from committing effects. An approved effect is applied by the state-transition machinery. Revision or rejection leaves the requested material effects unapplied, while the attempt can remain in the interaction history.

The resulting loop combines generative interpretation with state constraints (Table 2). Ownership, quantity, location, and duration are represented explicitly. Social interpretation, including the meaning of an exchange or an accusation, remains model-mediated upstream. Declared partner acceptance is an input to validation; it is not itself proof of independently

obtained human consent. Natural-language institutions are not automatically formalized as executable invariants by these checks. Consequently, the coordinator is a practical adjudication mechanism, not a proof of correct interpretation or universally valid social norms. Our evaluation reports observed outcomes rather than claiming exhaustive rule compliance.

4 Evaluation Design

We separate generated worlds, interaction events, action results, and participant judgments as different units of evidence. A world can pass structural checks without producing engaging play. An event can contain several effects or attempts, so the number of logged events need not equal the number of adjudicated results. Longer trajectories describe the simulated population, while human judgments require a separate instrument and participants.

The studies address four questions: **RQ1**, how reliably can a premise produce a complete world and at what cost; **RQ2**, which generated and open action pathways are actually used; **RQ3**, whether humans are reachable through the common interaction interface; and **RQ4**, what repeated social structure appears in longer historical trajectories. A fifth question asks whether high execution success coincides with material state changes under a longer, scripted policy harness. A sixth asks how the visual artifacts from seven model configurations are judged under a shared pipeline and premise. Supplementary structure controls vary the generation procedure within the same system and do not answer that cross-model question. Table 3 states the scope of each answer.

5 Construction Reliability and Internal Procedure Control

5.1 Paired Protocol and Completion

The generation comparison is an internal procedure control using the same Agora world representation. It uses Gemini 3 Flash Preview at temperature 0.2 and low thinking level for both methods; the single-pass condition is not an independently developed world-generation system. Five primary premises have three repetitions each; five held-out briefs have one each, giving 20 trials per method. The held-out set and evaluator were frozen before held-out generation. The requests include multi-sentence design constraints; this is not a strict

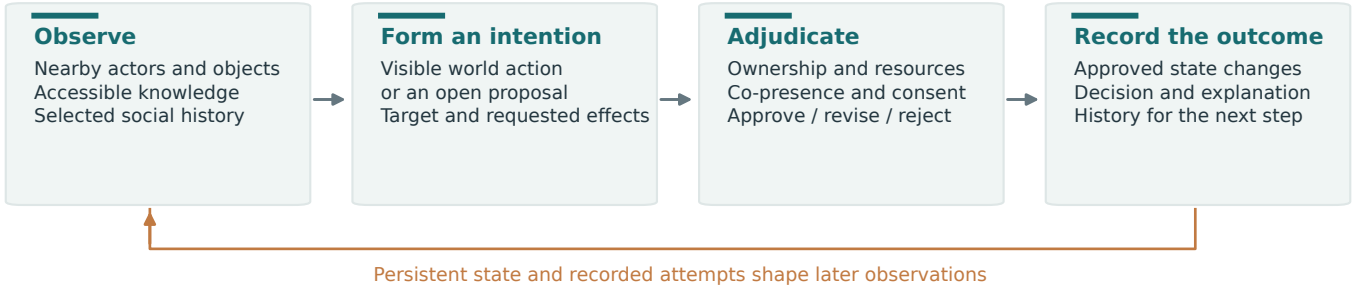


Figure 3. One interaction cycle. Bounded observations inform a visible action or an open proposal. The coordinator evaluates the proposed effects; approved changes update shared state. Both outcomes and failed attempts provide history for later decisions.

Study	Unit and scope	Evidence reported	Interpretation
Internal generation comparison	10 paired premises; 20 trials per method	Completion, structural quality proxies, cost	Descriptive reliability–cost tradeoff
Validation and repair	58 injected faults with clean checks; 5 repair cases	Detection, attribution, local repair cost	Specified contracts and seeded defects
Multiworld interaction	5 worlds; 2 rounds each; 53 events	Action pathways, dyads, outcomes, state changes	Initial behavior under common runtime settings
Cross-model audit	24 generations; 21 scripted 24-round runs	Completion, executed effects, rejected actions	Retrospective coverage and execution stress test
Historical pilot	3 prior runs; 1,052 events	Repeated ties and relational trajectories	Exploratory, confounded across worlds
Human model comparison	10 eligible returns; 40 image comparisons; 7 configurations	Exploration, map readability, character fit	Fixed pipeline and premise; sparse artifact preferences
Internal procedure control	20 structure comparisons; 10 requests	Same-system generation variants; full counts in appendix	Unequal budgets; no external-system comparison
Memory intervention	Planned 3 seed pairs, 32 rounds	Protocol only; no paired results here	Prospective within-world mechanism test

Table 3. Evaluation scope and evidence status. Completed observations are separated from evaluation materials and planned experiments.

one-sentence-input test (Appendix B). The analysis unit is the shared brief: primary-world binary outcomes use a majority over the three repetitions, while held-out outcomes use the single trial.

The single-pass baseline receives a 24,576-token output cap, three times the 8,192-token cap of one specialized stage. This provides a strong single-call allowance, but does not equalize total computation across methods. The compositional pipeline makes multiple calls and can use local retries. We therefore report first-pass and eventual completion separately, retaining failed outputs. The comparison evaluates the two complete generation procedures; it does not isolate decomposition at equal total token budget. This joint reporting of reliability and cost follows the evaluation concerns in Kapoor et al. [25].

Compositional generation completes 10/10 worlds versus 7/10 for the baseline; first-pass completion is 9/10 versus 5/10 (Table 4). The paired eventual outcomes contain three compositional-only successes and no reverse discordances; the one-sided exact paired test gives $p = 0.125$. For first-pass completion there are five compositional-only and one baseline-only successes ($p = 0.1094$). These small samples support a descriptive improvement, not a statistically established general advantage. Median generation is approximately 1.8–1.9 times faster for the single-pass method, which uses about half as many tokens.

World-level result	Compositional	Single-pass
First-pass completion	9/10	5/10
Eventual completion	10/10	7/10
Primary mean tokens/trial	48,527	22,250
Held-out mean tokens/trial	77,556	38,922
Primary median latency (s)	101.2	55.8
Held-out median latency (s)	161.1	86.1

Table 4. Completion is aggregated at the premise level; cost and latency are trial-level summaries within each split. The primary split has 15 trials per method and the held-out split has five.

5.2 Quality Proxies, Validation, and Repair

The six frozen quality measures show a mixed profile. Compositional outputs have higher mean merchant-inventory validity, premise-item recall, role uniqueness, room-canon adherence, and wardrobe-canon term coverage. The single-pass baseline has higher premise-agent recall (0.5593 versus 0.4632). The wardrobe proxy favors composition in nine of ten worlds (one-sided sign test $p = 0.0107$, unadjusted across the six measures). These are structural or lexical proxies; a higher score does not establish aesthetic preference or better play. We show all six in Figure 4, including the unfavorable comparison.



Figure 4. Construction evidence at its measured resolution. (A) Paired premise-level completion; the primary split uses majority outcomes. (B) Per-trial token use by split; Table 4 reports latency. (C) Differences in structural and lexical quality proxies, including the metric favoring the baseline. (D) Local-repair resource ratios for five cases. All values come from frozen records; no new model trials were run for this revision.

The seeded-fault suite mutates copies of five completed artifacts and invokes the production *deterministic validators*, without language-model or image-generation calls. It detects and attributes all 58 injected faults, with no flags in 58 corresponding clean checks. Faults cover missing semantic records, inventories, provenance mismatches, and image/assembly contracts. The clean checks reuse artifacts across categories, so they are not 58 independent worlds. This result concerns the specified fault suite; it is not 58 successful aesthetic judgments or evidence of universal defect detection.

For five injected role-inventory inconsistencies, localized refinement restores all five cases while preserving five unaffected intermediate stages. Mean model-call and token reductions relative to the full-rebuild reference are 48.0% and 39.9%. These observations support repair locality on the seeded failure type. A wider sample of naturally occurring defects is needed to estimate its general benefit.

5.3 Visual and Human-Evaluation Evidence

The five demonstration worlds connect generated maps, characters, objects, and persistent interaction. The Tidal Embassy trace includes rejected room images, local refinement, directional character atlases, and a 25-character interactive trial with movement, messaging, item use, trade, and a model-mediated

reply (Appendix G). This is implementation and case evidence, not a blinded comparison of visual quality.

The completed visual questionnaire evaluates preserved maps, character samples, and standardized world-structure diagrams. Section 6 reports the seven-configuration model comparisons. The separately collected structure judgments compare internal generation procedures and are retained in Appendix D.1. Neither arm measures actual play experience or cross-system superiority.

6 Human Model Comparisons Within Agora

6.1 Comparison Target and Included Responses

The human model comparison holds the Agora pipeline and promise-city premise fixed while varying the recorded language-model configuration. Participants see a preserved map and the same sampling of two character images per side, under masked A/B labels. The image generator is shared. Thus the measured objects are the resulting visual world artifacts under each configuration, not stand-alone LLM image generation and not competing world-authoring systems.

Ten of eleven returned English visual packets meet the released eligibility rule. V04 contains all eighteen choices but lacks cover confirmations and remains pending. The ten eligible returns supply forty image comparisons, covering all

21 pairs among seven configurations, and 120 item responses. Four pairs have one rater, sixteen have two, and one has four. Each comparison asks which world the respondent would rather explore, which map is easier to read, and which characters better fit their world. These criteria remain separate; they are not components of a validated overall-quality scale. The protocol and complete pairwise counts appear in Appendix D.

6.2 Observed Votes by Model Configuration

Table 5 adds a descriptive view of the model comparisons. For each configuration and question, we pool its observed wins and half-weighted ties and divide by its judgeable comparisons. Not sure choices are excluded from that denominator and remain in the full counts. This roll-up was added after responses were available to make the pairwise results readable; it is not a preregistered ranking. Opponent weights differ because the allocation and one cover exclusion leave unequal pair coverage.

Gemini 2.5 Flash receives eight of ten exploration preferences (80.0%) and seven of nine judgeable character-fit preferences (77.8%). GPT-5.6 Terra receives nine of thirteen exploration preferences (69.2%) and eleven of thirteen map-readability preferences (84.6%); GPT-5.6 Sol receives nine of eleven readability preferences (81.8%). The contrast within Sol is substantial: only four of eleven comparisons favor it for exploration. Gemini 3.1 Flash Lite receives six of ten exploration preferences but no readability wins and one readability tie (0.5/10). These are observations about fixed artifacts, not statistically established model superiority.

6.3 Criterion-Specific Pairwise Outcomes

Figure 5 reports all exact model pairs for exploration, map readability, and character-world fit. Each model name denotes the recorded world-generation configuration; the pipeline uses a shared image generator, so the comparison does not isolate an LLM’s ability to render pixels. Only one prompt has matched archived images.

The three criteria give different outcomes. Both raters prefer the GPT-5.6 Sol artifact to Gemini 3.5 Flash Lite for map readability, but both prefer the latter for exploration and character fit. The pairwise matrix preserves these distinctions instead of collapsing them into a single model ranking. Across the image arm there is one readability tie, one character-fit tie, and two character-fit abstentions. Four pairs now have one rater, sixteen have two, and one has four. This sparse, single-scene sample supports inspection of these particular outputs, not a stable ordering of model families or general claims about visual quality.

Separate internal control and unmeasured structure quality. The questionnaire also collected twenty comparisons between two generation procedures inside Agora. Their full results remain in Appendix D.1; the observed diagram preferences favor the single-pass procedure in aggregate. Both conditions share Agora’s representation, so these controls do not compare Agora with an independent system. They also do not measure structural differences *between the seven model configurations*. Human model comparisons here cover maps and characters; cross-model world-structure preference and actual play experience were not measured.

7 Interaction Across Five Generated Worlds

7.1 Protocol and Metric Definitions

The five held-out worlds are Aurora Court of Migrating Cities, Clockwork Rain Conservatory, Mycelium Patent Bazaar, Sunken Satellite Monastery, and Tidal Embassy of Lost Languages. Each contains 24–25 main characters and 11–12 discoverable actions spanning five interaction families. At least five activities per world derive from its premise.

Each accepted run uses Gemini 2.5 Flash for two rounds, activation probability 0.12, and a fixed world-specific seed (26080411–26080415). Agent-profile generation is not repeated during these runs. Image and video generation are disabled, and all five runs pass the world-identity audit. The runtime settings are shared, but the seeds and world contents differ: this is a descriptive multiworld study, not a paired intervention isolating premise effects.

An *event* is a recorded interaction. An observed participant is any actor or target ID, including the human interactor where present. A dyad is an unordered pair of distinct actor and target IDs. If n_{ij} is its event count, the repeat fraction is

$$\rho = \frac{\sum_{\{i,j\}} \max(n_{ij} - 1, 0)}{N_{\text{events}}}. \quad (5)$$

Action entropy is $H = -\sum_a p(a) \log_2 p(a)$ over logged route identifiers. An open event has a `proposed_route`; an explicit world-action event has a `world_action_route`. Other routes can still encode premise-specific activities. Thus these categories identify execution pathways, not an exhaustive semantic classification of world relevance.

Action-result success is computed separately from round summaries:

$$s = \frac{\sum_{w,t} n_{w,t}^{\text{success}}}{\sum_{w,t} n_{w,t}^{\text{results}}} = \frac{49}{56} = 87.5\%. \quad (6)$$

An event can include multiple effects or attempts. Neither 53 events nor 37 dyads is the denominator of this rate. Appendix A records the per-world numerators and denominators.

7.2 Action Pathways and Consequences

The runs produce 53 events, 37 within-world unique dyads, ten approved open proposals, and nine explicit world-action events (Table 6). Sunken Monastery uses eight explicit world actions, including purging corrupt packets and negotiating bandwidth tithes. Clockwork instead produces seven open-proposal events out of twelve. This contrast shows two available execution pathways in use, without proving that one causes better behavior.

In Clockwork, a jammed weather-sheet press becomes a local investigation. Marius confronts Valerius about the failure; Seraphina asks about etching; Valerius discusses sheet authenticity. In round two, the same participants revisit sabotage, inspect a brass folio, and redirect suspicion. Six actions form this three-character sequence; a seventh open proposal concerns the market impact of the jam. The approved effects update directed relationships while the press failure remains part of shared context. The complete episode diagram is retained in Appendix F.

Recorded configuration	Explore	Map readability	Character fit
Gemini 2.5 Flash	8/10 (80.0%)	7/10 (70.0%)	7/9 (77.8%)
Gemini 3.1 Flash Lite	6/10 (60.0%)	0.5/10 (5.0%)	4/9 (44.4%)
Gemini 3.1 Pro Preview	2/11 (18.2%)	4.5/11 (40.9%)	3.5/10 (35.0%)
Gemini 3.5 Flash Lite	6/14 (42.9%)	2/14 (14.3%)	6.5/14 (46.4%)
Gemini 3.7 Flash	5/11 (45.5%)	6/11 (54.5%)	5/11 (45.5%)
GPT-5.6 Sol	4/11 (36.4%)	9/11 (81.8%)	5/11 (45.5%)
GPT-5.6 Terra	9/13 (69.2%)	11/13 (84.6%)	7/12 (58.3%)

Table 5. Observed model-comparison votes on the single promise-city scene. Each cell is (wins + half ties) / judgeable comparisons, followed by its percentage. Not sure is excluded; half-integer numerators indicate a tie. Rows retain recorded model identifiers in fixed order. Models have unequal opponent weights, so these descriptive scores are not a balanced tournament, a general capability ranking, or independent samples across rows.

World	Events	IDs	Dyads	Repeat	H (bits)	Open	World route	Human-targeted
Aurora Court	9	12	7	22.2%	1.447	1	1	2
Clockwork Conservatory	12	8	6	50.0%	3.252	7	0	0
Mycelium Bazaar	9	12	7	22.2%	1.392	0	0	4
Sunken Monastery	11	13	9	18.2%	1.823	1	8	3
Tidal Embassy	12	8	8	33.3%	1.208	1	0	4
Total	53	–	37	–	–	10	9	13

Table 6. Two-round interaction records. IDs count observed participants, including the human where applicable. Open and world-route counts follow logged route prefixes; other routes may also be premise-specific. Dyads are unique within each world.

The resulting state changes differ across worlds. Clockwork accumulates +3 net trust and +21 influence/fear, whereas Aurora and Tidal accumulate +70 and +88 trust with zero net influence/fear. These are simulated state variables produced through the model-mediated interaction machinery. They show that the traces have persistent consequences, but they are not independent psychological measurements or evidence that the premise alone caused the differences.

7.3 Human Reachability and Failed Attempts

Thirteen events target the human interactor across four worlds; Clockwork has none. The logs establish that agents can select the player as a partner within the shared world. They do not establish how often a human replied, whether cooperation succeeded, or whether participants enjoyed the encounters. Those questions require participant-level outcome measures.

Seven of the 56 adjudicated action results are unsuccessful. Retaining these attempts in the history permits later revision, but this study does not measure whether failure feedback improves subsequent success. The evidence supports the availability of an outcome-sensitive interaction loop, consistent with the broader importance of environment feedback in language agents [17–19].

8 Cross-Model Construction and Scripted Execution

8.1 A Separate One-Sentence Corpus

The August 25 archive adds eight recorded model identifiers and three shared one-sentence prompts: a city taxing spoken promises by weight, a railway greenhouse allocating access through crop-care obligations, and a courthouse rebuilding a

shoreline from confiscated objects. This later pipeline requests 12 agents and four player slots. We re-audit its 24 generation outcomes separately from the earlier ten-brief comparison: the prompt set, implementation, and available models changed, so their scores are not pooled.

Seventeen of 24 trials yield complete artifacts, with nine first-pass completions. Five models complete all three prompts; Gemini 3.5 Flash Lite and GPT-5.6 Terra each complete only the first, and GPT-5.6 Luna completes none. Table 8 retains every failure. These are recorded API identifiers, not guarantees of immutable provider snapshots. One generation per model–prompt cell supports a coverage diagnosis, not a stable model ranking. Although the archive calls the suite hidden/confirmatory, it lacks a pre-execution immutable registration, and some scoring and retry policies were revised after development runs. We treat it as retrospective evidence.

8.2 What the Longer Harness Actually Measures

Seven first-prompt artifacts support three repeated 24-round trajectories each. The harness makes six calls per trajectory, at rounds 1, 5, 9, 13, 17, and 21; each call plans 12 actions over four rounds. Its planner observes summaries of *all* agents and recent events. It is a centralized multi-character policy, distinct from the locally observed agents in Section 3. Scripted requests, an invalid transfer, a resource shock, and a cross-room request probe execution. The prompt explicitly asks for thread continuation, legal recovery, and human follow-through; these behaviors cannot be claimed as spontaneous emergence or participant engagement.

The 21 trajectories contain 1,512 attempted model actions, of which 1,365 succeed (90.3%). Among successful actions, 398 (29.2%) record an inventory or location mutation. Among

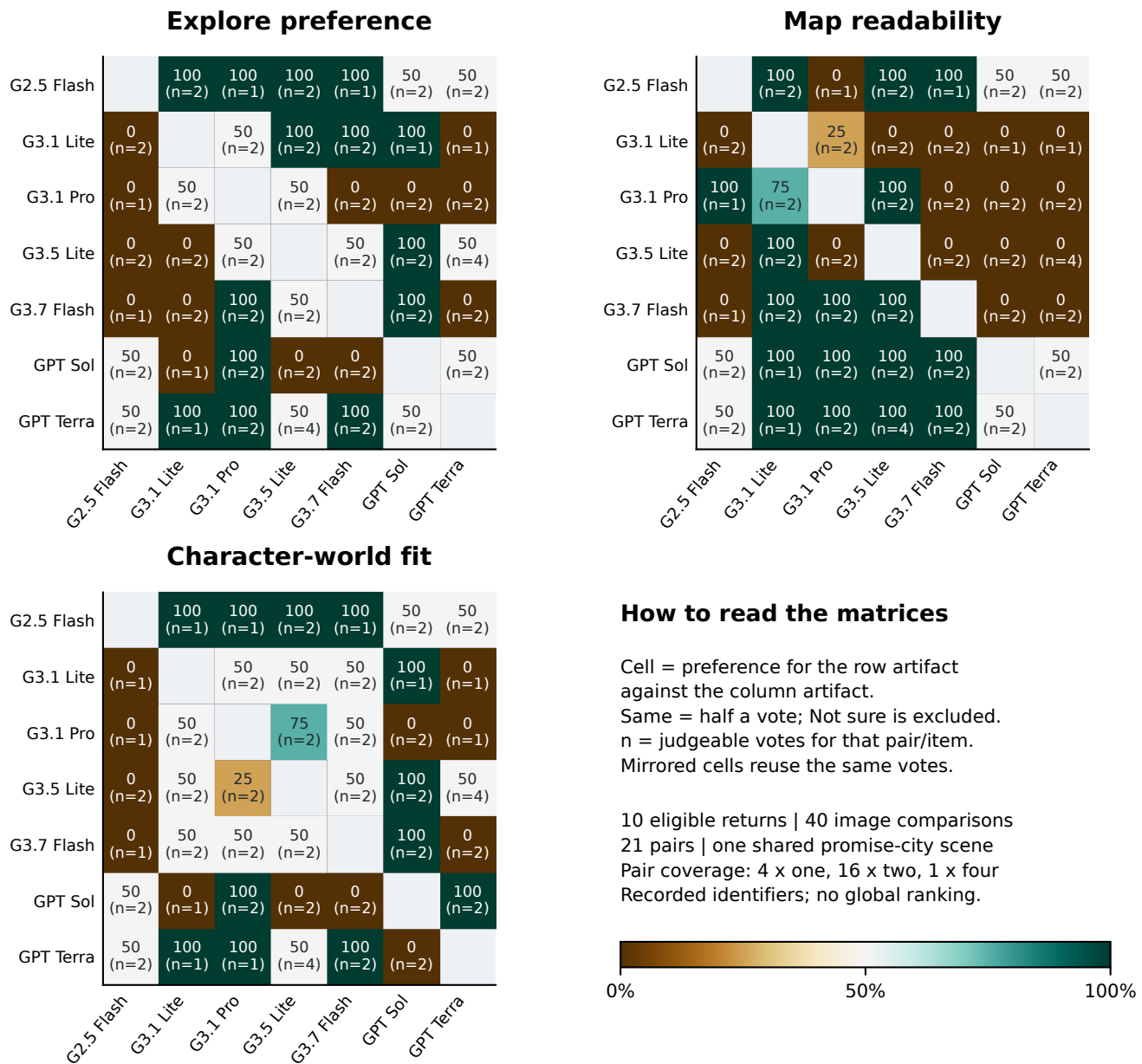


Figure 5. Images: all 21 pairs, with separate evaluation criteria. Cells report the percentage preferring the row artifact over the column artifact, counting ties as half votes; n is the number of judgeable votes. Not sure is outside the denominator. Mirrored cells describe the same comparisons, not additional observations. All outputs share one promise-city premise and the image pipeline; labels abbreviate archived backend identifiers. Exact counts, including abstentions, are in Appendix D.

314 approved proposals, 150 (47.8%) declare at least one non-speech effect; status-only effects count in this latter measure. These denominators address different questions and are not interchangeable. Figure 7 reports model-specific counts and repeat ranges instead of a weighted quality score.

8.3 Failure Modes and Evidence Boundaries

Gemini 3.7 Flash succeeds on all 216 attempts, but only 19 successes change inventory or location and only four of 22 approved proposals have non-speech effects. Terra’s 18 approved proposals are all speech-only, despite 214/216 successful actions. Conversely, Gemini 2.5 Flash has 42 failed attempts but 34/45 approved proposals with non-speech effects. These ob-

servations motivate reporting *what changed* alongside whether an action executed, consistent with analytical trajectory evaluation [24].

All 147 failed model actions and all 21 scripted invalid transfers leave their recorded state hashes unchanged. Final-state serialization round trips match in 21/21 runs; this checks serialization, not recovery after a process crash. Of 1,021 recorded relationship updates, 1,013 have the same $(\Delta\text{trust}, \Delta\text{affection}, \Delta\text{fear}) = (4, 2, 0)$. The custom-action handler supplies exactly this default for non-aggressive actions. Thus rising relationship values in this harness largely reflect execution rules rather than independently demonstrated social development. Appendix B provides the full audit and replica-

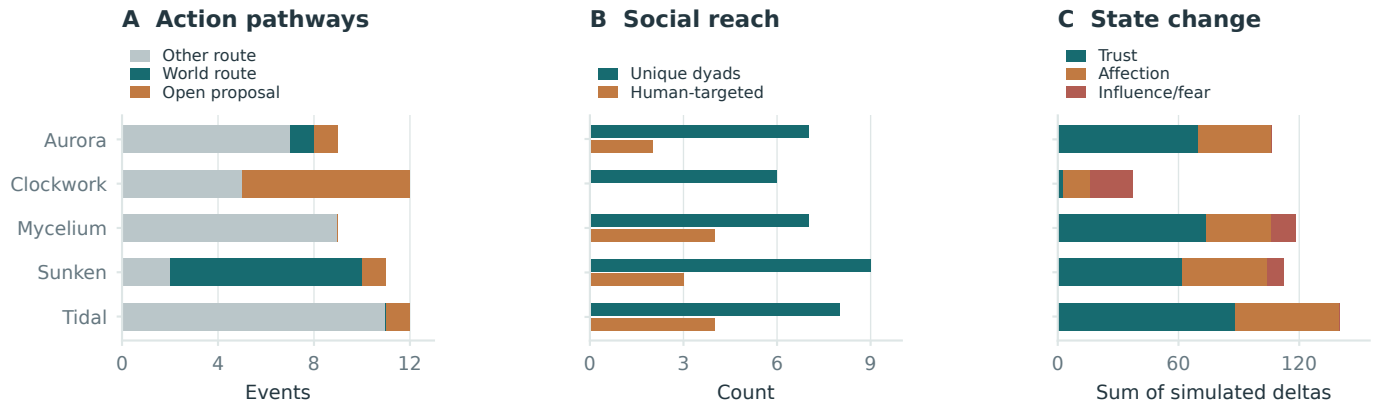


Figure 6. Interaction profiles under common runtime settings. (A) Logged action pathways. (B) Unique dyads and events targeting the human interactor. (C) Sums of model-mediated relationship changes. The five worlds have one short run each; bars are descriptive totals, not population estimates.

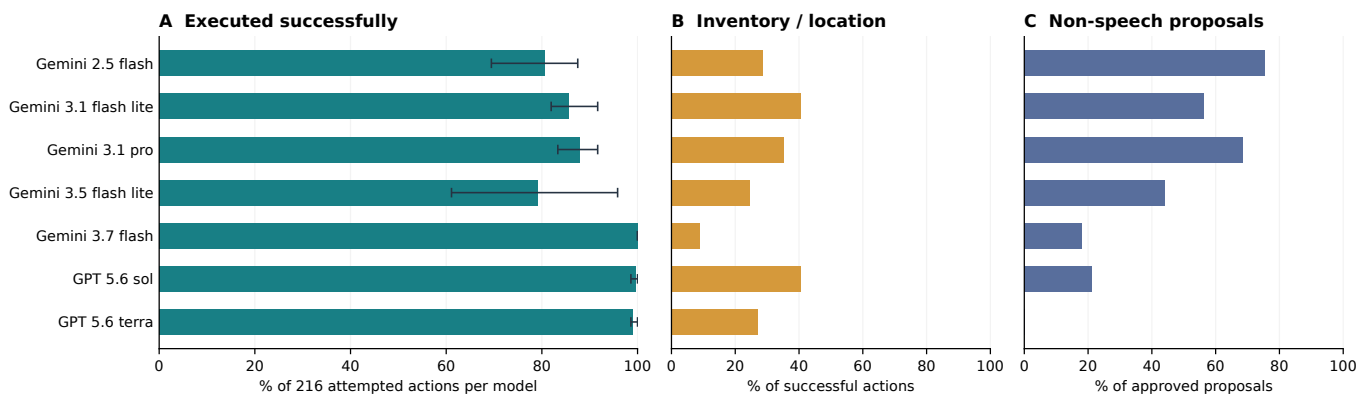


Figure 7. Successful execution does not ensure rich consequences. Seven generated worlds, each with three 24-round centralized-policy runs. (A) Success per attempted action; whiskers are the minimum and maximum over three runs, not confidence intervals. (B) Inventory/location mutations among successful actions. (C) Non-speech effects among approved proposals, including status changes. All comparisons couple a model with its own generated world; they do not isolate policy capability on identical environments.

tion boundaries.

9 Historical Context

Three earlier runs add 1,052 events across an adventuring guild, a black-market auction, and a storm-struck cruise. Between 23.4% and 36.5% of events revisit an existing dyad, while the largest connected components contain only 11–13 participants. These records motivate studying communication across local clusters, but configurations differ and spatial scheduling can produce the observed structure. Full tables and trajectories appear in Appendix C; they are not pooled with the later scripted harness.

10 Discussion and Limitations

A contribution at the connection between authoring and simulation. Agora connects staged world authoring, explicit persistent state, and grounded social action. Its closest predecessors already establish language-driven game construction [3, 4], shared human–agent play [12], and Game Master adjudication [6]. The contribution is their integration around a premise-conditioned world representation and inspectable trajectories. The model comparison varies language-model configurations within Agora. The generation controls vary its in-

ternal authoring procedure. Neither compares Agora against those independent systems; no cross-system superiority is established.

Discoverability and expressiveness are separate design needs. A generated repertoire explains what inhabitants can do in a new setting. Open proposals let a particular encounter depart from that vocabulary. The Sunken and Clockwork traces illustrate both pathways. Their presence does not establish the benefit of openness: a matched proposal ablation, with equivalent world and agent states, is needed to estimate that effect. Likewise, interface access does not establish useful or enjoyable human collaboration.

Internal procedure controls have limited scope. Ten shared briefs compare staged and single-pass generation under the same representation, with different total computation and retry opportunities. The observed completion gains characterize that implementation comparison, not the value of Agora’s shared representation or its superiority to an external system. The supplementary diagram judgments favor single-pass outputs in aggregate and remain reported in full. Both favorable and unfavorable control outcomes have this same restricted scope.

The visual study is narrow and exploratory. Ten eligible coded returns supply forty model-image comparisons; a separate arm supplies twenty internal structure controls. One of eleven returns lacks cover confirmations. The image arm covers one scene, with one to four ratings per exact pair, and the structure arm shows containment diagrams rather than a complete interactive world. The source identities are masked, but the artifacts share a generation pipeline and retain unequal compute. The organizer reports unpaid online volunteer recruitment; an ethics approval or exemption record and completion times remain undocumented. Accordingly, the findings characterize these returned artifact judgments, not a stable model ranking or human enjoyment.

Execution is not behavioral validation. The five locally observed runs last only two rounds with one seed each. The 21 longer runs use centralized block planning, scripted human events, and each model’s own generated world. Their high execution success does not validate emergent coordination; identical relationship increments expose a contribution from default state-update rules. Historical worlds are confounded by prior configurations. Relation deltas are model-mediated state variables, and route entropy depends on the action vocabulary. Neither measures human trust, fear, affection, or social intelligence directly. LLM judging also carries systematic biases [23]; behavioral fidelity would require independent human data and explicit validation targets, as in human-grounded agent studies [27].

A prospective intervention is not a completed result. The Tidal protocol varies access to prior social episodes while preserving current world state. No completed paired 32-round results or blinded annotations are reported in this release. The available record contains a local protocol, but no external registration identifier; we therefore describe it as a prospective design. Three seed pairs would provide a feasibility estimate with substantial uncertainty, not precise population inference.

Reproducibility and human participation. Hosted models may change despite a recorded model name; available settings, seeds, exclusions, and frozen evidence are documented in Appendix A. Human studies should evaluate refusal, outcome legibility, and collaboration under unequal information. Fictional roles do not validate demographic models.

11 Conclusion

Agora turns a short premise into a rendered social environment whose inhabitants can act on persistent shared state. The completed studies show criterion-specific visual preferences across seven model configurations, a reliability–cost tradeoff in an internal generation control, use of explicit world actions and open proposals, and a gap between successful execution and material consequences in a separate 21-trajectory audit. The central result is an inspectable connection between world authoring and consequential interaction. Broader human evaluation and controlled interventions are needed to determine how well the generated worlds support collaboration and which mechanisms shape their longer social trajectories.

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World	Successes	Results	Rate
Aurora	9	9	100.0%
Clockwork	11	13	84.6%
Mycelium	8	10	80.0%
Sunken	11	11	100.0%
Tidal	10	13	76.9%
Total	49	56	87.5%

Table 7. Audited action-result counts, summed directly from the accepted story bundles’ round summaries. The 61-result denominator in the previous manuscript was a transcription error.

A Reproducibility and Evidence Audit

Generation settings and aggregation. The frozen protocol uses `gemini-3-flash-preview`, temperature 0.2, and low thinking level. Specialized stages have an 8,192-token output cap; the monolithic call has a 24,576-token cap. Five primary premises have three repetitions each and five held-out premises have one each. A primary premise passes a binary completion endpoint when a majority of its repetitions pass; the five held-out premises use their single observed outcome. Costs summarize all trials within each split. Failed outputs are retained. Quality-evaluator definitions and the held-out set were frozen before held-out generation.

The manuscript’s construction figures read the frozen file `paper_evidence_summary_20260729.json`. The six quality proxies are merchant-inventory pass rate, premise-item recall, premise-agent recall, role uniqueness, room visual-canon adherence, and wardrobe-canon term coverage. They are reported as proxies and are not substituted for human judgments. The seeded validator report is included alongside these summaries.

Accepted interaction runs. The accepted records are `mw3_aurora`, `mw3_clockwork`, `mw3_mycelium`, `mw3_sunken`, and `mw3_tidal`, collected on August 4, 2026. They use Gemini 2.5 Flash through AI Studio, two rounds, activation probability 0.12, and seeds 26080411–26080415 in that order. Compiled profiles are reused, and image/video generation is disabled. The archived run configuration does not by itself establish an immutable provider model snapshot; decoding details absent from that record are not inferred.

Earlier `mw_` attempts are excluded because a shared mutable scenario directory caused cross-world contamination. Earlier `mw2_` attempts made no model progress because DNS was unavailable. The accepted runs place mutable scenario state in separate directories and pass the world-identity audit. These exclusions concern execution validity and are distinct from excluding an unfavorable behavioral outcome.

Success-count correction. Table 7 distinguishes action results from story events. The frozen metrics already reported 0.875; summing the raw round summaries recovers 49 successes from 56 results. This revision corrects the prose denominator without changing the recorded success rate or event count. The accompanying audit file includes the source paths, SHA-256 hashes, and extracted round counts so that this correction is traceable.

Historical analysis. The pilot records are the guild run of April 22, 2026, the black-market run of April 19, and the cruise run of April 18. They are analyzed independently of the August interaction runs and the planned intervention. Observed-ID counts are 103, 99, and 100, respectively; these are counts of recorded actor/target identifiers, not a claim that all runs had an identical realized population.

Artifact scope. The release contains the paper sources, BibTeX database, figure programs, frozen figure inputs, evidence-audit summaries, and the English visual survey returns with their fixed allocation, extracted choices, completed Excel workbook, and descriptive-analysis script. The bundled inputs reproduce the figures and reported summary arithmetic; full runtime replication additionally requires the original implementation, configuration, model access, and raw execution records. The expanded release reanalyzes existing August 25 generation and execution records. This revision incorporates eleven returned survey files, ten eligible for analysis. No new model call or paired memory-intervention result is claimed. The figure and table inputs retain the response exclusions and every Not sure choice, so the published summaries can be re-generated.

B Extended Evidence: Counts and Harness Audit

Source selection. Generation outcomes come from the 24 result records under the archive’s `confirmatory_runs` directory. A first-pass completion is a complete artifact with no recorded localized generation retry. Retry lists for some failed cases were not preserved in the original format; we therefore do not estimate total repair incidence from those lists. Failed cases remain failed. The 21 execution sources are the corrected `runtime_trajectories_v2/replicate_*` records. Earlier runs affected by human placement and transport defects are not substituted or pooled. All raw result and trajectory records used here are frozen with SHA-256 hashes and recomputed by the bundled audit script.

Input scope of the earlier construction comparison. The ten requests in Section 5 are *world briefs*: some include several sentences specifying institutions, cast size, room requirements, and visual style. They do not test strictly one-sentence input. For the comparative questionnaire, the archived single-pass prompt contains an original-request JSON exactly matching the corresponding compositional request for all ten selected first replicates. This confirms a shared user request; the method-specific system instructions and total computation differ. The three later prompts are separate, genuinely one-sentence semantic inputs.

Execution schedule and observability. Each four-round block is planned from one snapshot; intermediate actions can make later planned actions invalid. At each checkpoint the policy receives room connectivity, all-agent summaries, the action catalog, up to four open action examples, the current intervention, and the 24 most recent events. Decoding uses the recorded temperature 0.75 and low thinking setting. There is no archived claim of matched random seeds across providers. A model generates and controls its own world, so world difficulty, genera-

Recorded model	Complete	First pass	Executed	Inventory/location	Approved	Non-speech
Gemini 2.5 Flash	3/3	2/3	174/216	50/174	45	34/45
Gemini 3.1 Flash Lite	3/3	2/3	185/216	75/185	89	50/89
Gemini 3.1 Pro Preview	3/3	0/3	190/216	67/190	19	13/19
Gemini 3.5 Flash Lite	1/3	1/3	171/216	42/171	102	45/102
Gemini 3.7 Flash	3/3	3/3	216/216	19/216	22	4/22
GPT-5.6 Luna	0/3	0/3	N/E	N/E	N/E	N/E
GPT-5.6 Sol	3/3	1/3	215/216	87/215	19	4/19
GPT-5.6 Terra	1/3	0/3	214/216	58/214	18	0/18
Total	17/24	9/24	1365/1512	398/1365	314	150/314

Table 8. Retrospective counts without reviewer scores or rank ordering. Construction columns use three prompts per model. Execution columns use only the first prompt’s generated artifact and three repeated trajectories. N/E means no eligible artifact; it is not a runtime zero or a human preference. “Non-speech” includes status effects and is counted from approved proposals.

tion quality, and policy quality are confounded. The harness is an executable-plan stress test, not an information-asymmetric social experiment.

Scripted human and recovery probes. The round-5 request and round-21 follow-up are supplied by the harness. When needed, the harness moves the human into an occupied room before a request or rejection probe; this movement is not counted among model actions. The round-9 probe requests transfer of an item the human does not possess. All 21 such probes are rejected without a state-hash change. A resource shock at round 13 and a cross-room instruction at round 17 are likewise scripted. Target acceptance in model-generated proposals is a declared field, not an independently elicited human response.

What counts as a consequence. The inventory/location endpoint includes successful movement as well as inventory effects, and therefore is not a measure of economic transformation alone. The non-speech endpoint includes temporary statuses and describes the requested effects of approved proposals; approval does not establish that every intended semantic consequence occurred. The 147 failed model actions contain both before/after hashes and a false state-change flag. Five references to nonexistent event IDs occur in failed Gemini 2.5 Flash actions; no successful action in the archive has such an unknown reference. Thread labels and event references are supplied by the planner, so retaining them is not independent proof of causal coherence.

Statistics and reproducibility. The independent scope is 24 generation cells and 21 trajectory records, not 1,512 independent experimental units. We report counts and the observed range over three runs, without asymptotic confidence intervals, significance stars, or a composite leaderboard. The audit reads raw outcomes and mutations and does not load the identity-visible expert review or its weighted scores. The source snapshot also includes the planner harness, coordinator validation, and custom-action handler used to interpret the records. Source inspection supports the mechanism explanation but does not replace fresh execution tests against an immutable historical checkout.

C Exploratory Historical Trajectories

We separately analyze three prior runs configured around approximately 100 agents: an adventuring guild (25 rounds), an underground black-market auction (20), and a storm-struck luxury cruise (20). They predate the current hypotheses and differ in configuration and social context. We reconstruct undirected actor–target networks, route entropy, and sums of directed relationship changes using the same metric definitions.

Between 23.4% and 36.5% of events revisit an existing dyad, while each world contains 229–282 unique dyads (Table 9). Route entropies are 2.889, 2.619, and 2.524 bits for guild, black market, and cruise, respectively. Because the repertoires differ, these values describe each trace’s variety; they are not directly comparable measures of social intelligence.

The largest components contain only 11–13 participants despite many observed IDs. Figure 8 shows dyads continuing to accumulate after component sizes stabilize. This is consistent with additional ties forming predominantly within local clusters. Spatial encounter and activation policies can generate such structure, so its social interpretation must remain conditional on those mechanisms.

In the black market, influence/fear (+1194) exceeds trust (+179) and affection (+9); the guild and cruise instead accumulate mostly trust and affection. The pattern is compatible with each scenario’s institutions, but prompts, action definitions, schedules, and run histories are confounded. It motivates a controlled question rather than answering one. In particular, warning diffusion and collective rescue need separate outcomes: receiving information need not produce action [28]. Appendix E specifies a paired memory intervention to examine that distinction.

D Returned Visual Survey: Protocol and Complete Counts

Version and evidence. This model-focused presentation retains both originally collected arms and all eligibility rules; no responses or unfavorable internal-control outcomes are removed. The per-model pooled summary was added after collection. This analysis uses the seven-page, all-English visual packets V01–V11, released September 22, 2026, and the returned archive supplied on GitHub before this revision.

World	Rounds	Events	IDs	Dyads	Repeat	Largest comp.	Dominant state changes
Guild	25	444	103	282	36.5%	11	Trust +3473; affection +1745
Black market	20	299	99	229	23.4%	13	Influence/fear +1194
Storm cruise	20	309	100	235	23.9%	13	Trust +2576; affection +961

Table 9. Exploratory statistics from 1,052 historical events. IDs are distinct logged actor/target identifiers, not a fixed population denominator; the guild record contains 103. Relationship totals are simulated state changes.

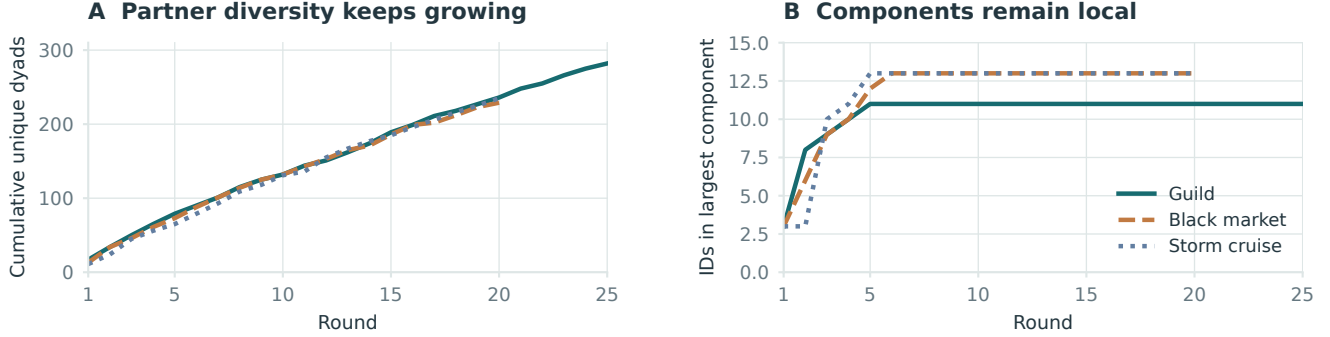


Figure 8. Longer traces distinguish activity from connectivity. Unique dyads continue to accumulate, while the largest observed components plateau early. These historical runs motivate testing transmission across local clusters; they do not identify its causal mechanism.

It supersedes the text-based prospective evaluation in earlier manuscripts; older Q/S/P instruments are not pooled with these responses. Each coded return has six comparisons and eighteen answers. The source PDF form values are checked against the selected widget appearance states before decoding through the frozen allocation. No answers are inferred from a model, imputed, or edited. The filled Excel file preserves every return, with V04 marked for cover verification and omitted from all preference summaries.

Visual sampling. The image arm uses the one archived premise with complete matching render assets: a cliffside city that taxes spoken promises by weight. Each of seven world configurations contributes its unaltered map and the first idle-down frame of its first two requested characters, under the same framing rule. These are finalized pipeline assets, including recorded repair steps, not independent fresh generations. The unavailable Luna artifact is not replaced or counted as a preference loss. Terra’s unavailable third publishable character is not shown; no condition is judged on a third character. This instrument does not evaluate asset completeness, animation, or navigation.

Structure sampling and presentation. The ten matched requests use the fixed first archived replicate of full Agora and single-pass generation. The diagrams include all source rooms in source order, count inhabitants by their recorded home location, and show the count of large scene components with the first component label as an example. Lines denote containment, not adjacency. The same short English scene description appears on both sides. The comparison retains readable source records even when an objective completion contract fails. There are no matched baseline renderings: no unrelated or synthetic image is substituted. The conditions retain unequal total generation budgets.

Allocation and masking. Seed 20260922 fixes the schedule before response analysis. Every participant sees four distinct image pairs and two distinct structure pairs. The original plan has 44 image and 22 structure comparisons; each exact pair occurs twice, with one seed-selected pair per arm occurring four times. Every pair has balanced A/B orientation, and Agora appears on A once and B once for each participant. Excluding V04 removes four image and two structure comparisons, leaving the coverage counts reported in Section 6. Individual pair orientation is consequently no longer universally balanced. The twenty retained structure comparisons remain balanced overall: Agora is shown on each side ten times. The questionnaire hides source names, but visible artifacts can still reveal stylistic clues; analyst blinding is not claimed.

Questions and estimands. Q1 in both arms asks, “Which world would you rather explore?” Image Q2 asks which map is easiest to read; Q3 asks which characters fit their world better. Structure Q2 asks which world is easiest to understand; Q3 asks which places fit the scene better. All offer A, B, Same, and Not sure. Exploration measures *stated interest from the supplied evidence*, not enjoyment after playing. The diagnostic items measure different constructs across the arms and are never merged. This explicit separation follows the reporting concerns of Howcroft et al. [29].

Let $w_{jq}, t_{jq}, \ell_{jq}$ denote Agora wins, ties, and baseline wins for request j and item q , after eligibility filtering. With abstentions excluded,

$$\hat{p}_{jq} = \frac{w_{jq} + t_{jq}/2}{w_{jq} + t_{jq} + \ell_{jq}}, \quad \hat{p}_q = \frac{1}{10} \sum_{j=1}^{10} \hat{p}_{jq}. \quad (7)$$

The ten-request mean is shown only if all ten denominators are nonzero. All three items meet this condition. Pooled vote scores, provided only as a weighting diagnostic, are 35.0%, 42.5%, and 36.8%, versus equal-request scores of 40.0%,

32.5%, and 37.5%. Clockwork has four votes and is preferred for clarity but not exploration; reporting the estimator explicitly prevents this imbalance from silently changing the result. Image scores use the same tie and abstention rule separately within each exact model pair, with no cross-prompt or global model ranking.

Eligibility, consent, and missing administration records. Ten covers affirm age 18 or older, English comprehension, voluntary participation, and no prior work on the system or exposure to detailed results. V04 has eighteen choices but no cover answers; blank checkboxes are treated as unconfirmed, not as an affirmative response or a known refusal. The exclusion rule was included in the original release and does not depend on preferred source. The archive supplies no names or demographics beyond the cover confirmations, nor independent verification that codes correspond to distinct people. The organizer reports online volunteer recruitment with no compensation, and identifies V04 as an accidentally omitted cover. This explanation does not supply the missing participant confirmations. No sampling frame or probability-sampling procedure is documented; recruitment is treated as an online volunteer sample. Actual completion time and an ethics approval or exemption record were not supplied. No approval, exemption, representative sampling, or measured 10–15-minute completion time is claimed.

Response dependence and presentation checks. The 180 item responses come from ten coded returns; they are not 180 independent people. All image comparisons reuse seven artifacts on one scene, and all three questions reuse each displayed pair. We report counts and request-level variation, without item-level significance tests, Elo scores, or confidence intervals that would imply unsupported population precision. Image Q1 selects A 15 times and B 25 times; structure Q1 selects A 11 times and B 9 times. This descriptive side asymmetry is retained for inspection, but does not identify a position effect separately from artifact preference. No respondent is excluded for a repeated choice pattern or for favoring the baseline. Very small per-pair counts make agreement coefficients unstable; the full counts are reported instead.

Image identity key. M1: Gemini 2.5 Flash; M2: Gemini 3.1 Flash Lite; M3: Gemini 3.1 Pro Preview; M4: Gemini 3.5 Flash Lite; M5: Gemini 3.7 Flash; M6: GPT-5.6 Sol; M7: GPT-5.6 Terra. These names preserve archived backend identifiers and do not independently verify product naming, an immutable provider snapshot, or common inference budgets. The actual visual artifacts, rather than the labels, are the measured objects.

D.1 Internal Generation Procedure Control

This supplementary comparison varies the generation procedure within the Agora implementation. Both conditions use its world representation and rendering conventions; they are not independent world-generation systems. The total generation budgets also differ, so this is a procedure control, not a compute-matched component ablation. The returned results below are retained in full and must not be interpreted as Agora’s performance against an external system. For each item, a pref-

erence for Agora scores 1, a tie 0.5, and a baseline preference 0; “Not sure” is excluded from that item’s preference denominator and reported separately. We average within each request and then equally across all ten requests, as specified in the released workbook. This prevents the request receiving four ratings from dominating the overall estimate.

Respondents select the baseline in thirteen of twenty exploration choices, versus seven for Agora. Agora’s equal-request preference is 40.0% for exploration, 32.5% for world clarity, and 37.5% for place–scene fit (Table 12). These observations favor the single-pass condition on the displayed containment diagrams. They neither benchmark Agora against a separate system nor measure the contribution of its shared representation, renderer, or runtime. A causal claim about decomposition would additionally require equalized budgets and controlled retries.

The pattern varies by request (Figure 9). Both Tidal raters prefer Agora for exploration. All four Clockwork raters prefer the baseline for exploration while choosing Agora for clarity. Understandability and desire to explore are thus distinguishable judgments in these returns; we do not sum them into a single quality score. One or two raters on most requests cannot identify why a design was preferred, and the diagram omits institutions, social behavior, and actual play. Equal-compute generation and interactive evaluation remain separate open tests.

E Prospective Tidal Memory Intervention

Status. This appendix describes planned work, not an additional completed experiment. The project contains a local protocol and a one-round full-memory smoke run; it does not contain completed paired 32-round results supporting a causal memory claim. No external preregistration identifier is available in the release. The changes to analysis and measurement specified below must be frozen before confirmatory runs and are not represented as an already registered plan.

E.1 Intervention and Hypothesis

The 25-agent Tidal Embassy world is initialized once and reused. At round zero, three agents in the lower archive privately learn that a flood threatens the only surviving peace treaty unless it is moved and authenticated. The full-memory condition retains prior social episodes. The ablation removes prior-round social episodes from decision context while preserving current locations, inventories, public facts, task state, and directed relationships. Thus it tests access to episodic social history, not removal of all memory: tasks, current knowledge, and relationships can also carry historical signals.

The planned seeds are 26080311, 26080312, and 26080313, with 32 rounds per condition. Initial profiles, world configuration, model and decoding settings, and activation schedule are matched within each pair. Activation draws should be precomputed with a separate random stream; reusing a seed alone does not guarantee matching if the conditions consume randomness differently. API nondeterminism remains a source of variation.

The hypothesis is that access to earlier social episodes increases warning diffusion and coordinated rescue within this generated setting. Generative Agents motivates studying memory in community behavior [5]; the separation of information

Request	Explore	Clarity	Scene fit
Archive Of Borrowed Gravity	50% (1/0/1/0)	25% (0/1/1/0)	50% (1/0/1/0)
Aurora Court Of Migrating Cities	50% (1/0/1/0)	100% (2/0/0/0)	100% (2/0/0/0)
Cartographer Lung Exchange	0% (0/0/1/0)	0% (0/0/1/0)	0% (0/0/1/0)
Clockwork Rain Conservatory	0% (0/0/4/0)	100% (4/0/0/0)	25% (1/0/3/0)
Intertidal Embassy For Extinct Rivers	0% (0/0/2/0)	0% (0/0/2/0)	0% (0/0/2/0)
Museum Of Future Debts	0% (0/0/2/0)	0% (0/0/2/0)	0% (0/0/2/0)
Mycelium Patent Bazaar	100% (1/0/0/0)	0% (0/0/1/0)	0% (0/0/1/0)
Night Market Of Unfinished Weather	50% (1/0/1/0)	50% (1/0/1/0)	50% (1/0/1/0)
Sunken Satellite Monastery	50% (1/0/1/0)	0% (0/0/2/0)	50% (1/0/1/0)
Tidal Embassy Of Lost Languages	100% (2/0/0/0)	50% (1/0/1/0)	100% (1/0/0/1)

Table 10. Every structure request. Each cell gives the Agora preference score and counts (Agora wins / ties / baseline wins / Not sure). The requests, not the number of ratings, are equally weighted in the main result.

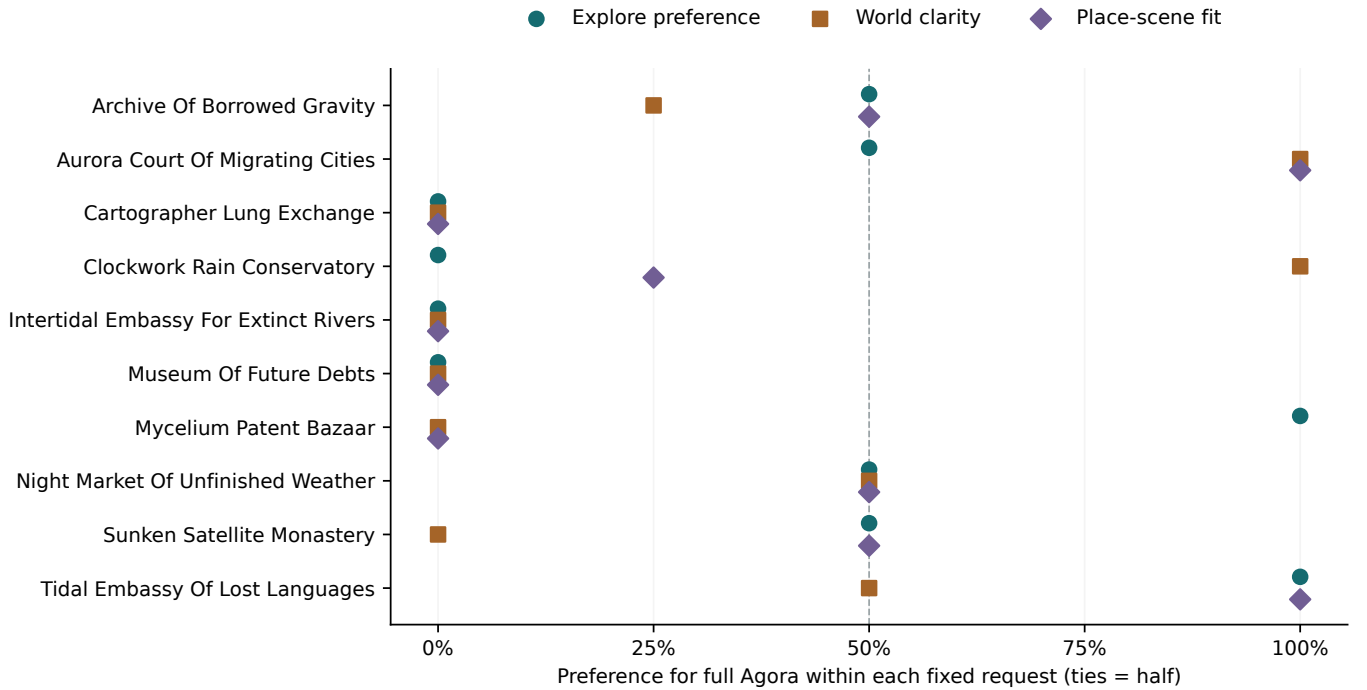


Figure 9. World structure: preferences vary by request and item. Each point is the fraction favoring full Agora for one request and question, with a half vote for a tie. The analysis includes 20 comparisons: seven requests have two raters, two have one, and Clockwork has four. For Tidal scene fit, one of two raters selects Not sure. These are descriptive fixed- artifact scores; the dashed line marks equal preference, not a significance threshold.

reach and collective action is also informed by contagion research [28].

E.2 Outcomes and Measurement Controls

The primary endpoint is warning reach by round 16 among the 22 initially uninformed agents. Let $Z_i(16)$ indicate that the agent has expressed or recorded a prespecified treaty-threat proposition by that round. Then

$$D_{16} = \frac{1}{22} \sum_{i \notin I_0} Z_i(16), \quad |I_0| = 3. \quad (8)$$

Measurement must use an append-only audit stream outside the ablated decision memory; otherwise deleting memory could mechanically reduce measured reach even after successful

transmission. The audit record is not fed back to agents. Initial warning availability must be preserved equally through its first intended use, so the ablation does not inadvertently remove the treatment trigger itself.

Secondary outcomes are first cross-room transmission time, rescue-task participation and role diversity, treaty relocation and authentication, unique and repeated dyads, largest component, object transfers, relationship changes, and rule violations. Unreached transmission or rescue events are reported as censored at the horizon. Two condition-blind annotators should label candidate episodes under a frozen proposition codebook; agreement and resolved disagreements must be reported before condition-level interpretation.

Manipulation checks verify that only the designated agents

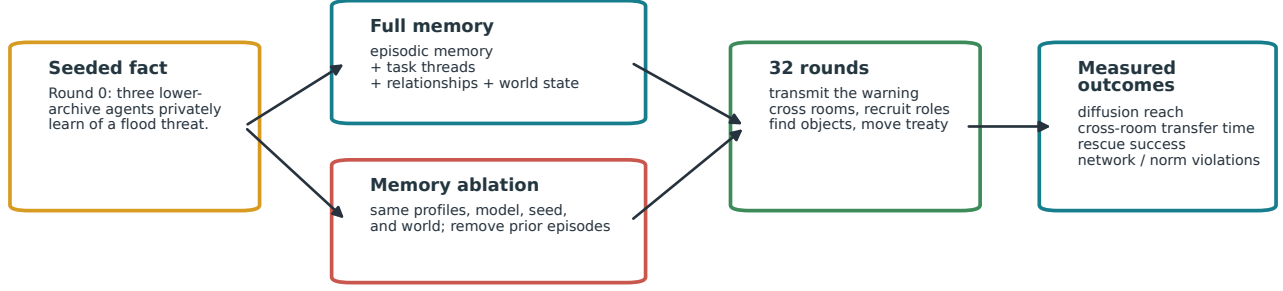


Figure 10. Prospective design; no paired results reported. Matched initial states receive the same private warning. Conditions differ in access to prior social episodes. Diffusion and material rescue are distinct outcomes. The planned unit of comparison is a seed pair.

Pair	Explore	Map	Characters
M1–M2	2/0/0/0	2/0/0/0	1/0/0/1
M1–M3	1/0/0/0	0/0/1/0	1/0/0/0
M1–M4	2/0/0/0	2/0/0/0	2/0/0/0
M1–M5	1/0/0/0	1/0/0/0	1/0/0/0
M1–M6	1/0/1/0	1/0/1/0	1/0/1/0
M1–M7	1/0/1/0	1/0/1/0	1/0/1/0
M2–M3	1/0/1/0	0/1/1/0	1/0/1/0
M2–M4	2/0/0/0	0/0/2/0	1/0/1/0
M2–M5	2/0/0/0	0/0/2/0	1/0/1/0
M2–M6	1/0/0/0	0/0/1/0	1/0/0/0
M2–M7	0/0/1/0	0/0/1/0	0/0/1/0
M3–M4	1/0/1/0	2/0/0/0	1/1/0/0
M3–M5	0/0/2/0	0/0/2/0	1/0/1/0
M3–M6	0/0/2/0	0/0/2/0	0/0/2/0
M3–M7	0/0/2/0	0/0/2/0	0/0/1/1
M4–M5	1/0/1/0	0/0/2/0	1/0/1/0
M4–M6	2/0/0/0	0/0/2/0	2/0/0/0
M4–M7	2/0/2/0	0/0/4/0	2/0/2/0
M5–M6	2/0/0/0	0/0/2/0	2/0/0/0
M5–M7	0/0/2/0	0/0/2/0	0/0/2/0
M6–M7	1/0/1/0	1/0/1/0	2/0/0/0

Table 11. Every image pair. Entries are first-source wins / ties / second-source wins / Not sure. Each row describes the same pair under three different questions; these are not independent respondent samples.

initially receive the warning, that paired initial states and activation schedules match, and that the ablation removes the intended history without exposing hidden facts through summaries or coordinator feedback. These checks implement the information-boundary concern highlighted by Zhou et al. [21].

E.3 Analysis and Precision

The independent comparison unit is a seed pair, not an event or an agent. For seed s , report $\Delta_s = D_{16,s}^{\text{full}} - D_{16,s}^{\text{ablation}}$, both condition trajectories, and the material rescue outcome. Report all three differences and their mean and range. Three pairs are too few for a stable bootstrap confidence interval or a well-powered confirmatory test; additional seeds and a prospectively justified

Item	Full	Base	Tie	?	Score
Explore	7	13	0	0	40.0%
Clarity	8	11	1	0	32.5%
Scene fit	7	12	0	1	37.5%

Table 12. Structure judgments from ten eligible returns. Full and Base denote Agora and the single-pass baseline, irrespective of randomized A/B labels; ? = Not sure. Counts pool twenty comparisons per item. Score is Agora’s *equal-request* mean, not a pooled vote fraction. No population superiority is inferred from these small counts.

sample size are needed for precise inference.

Qualitative selection rules are the earliest cross-room transmission, first multi-role rescue, largest trust reversal, and first failed coordination. Selections should be applied identically to both conditions and report when an event does not occur. The existing one-round smoke run is execution validation only and must remain outside these comparisons.

F Clockwork Episode

The diagram follows the short episode described in Section 7. It is an illustrative trace from one two-round run, not an independently replicated behavioral effect. Six actions form the central three-character sequence; the seventh concerns the market impact of the malfunction.

G Tidal Embassy: A Generation and Play Trace

We now follow *Tidal Embassy of Lost Languages* from premise to play. Its one-sentence premise describes a floating embassy where linguists, smugglers, ambassadors, and memory divers trade endangered languages as political assets. The example preserves intermediate images and refinement attempts, allowing each generative stage to be connected to the final playable world.

Step 1: Premise analysis and planning. The planning stage identifies the nouns that become state—embassy, languages, translation rights, archives, smugglers, diplomats, and tides—and the verbs that imply gameplay—trade, translate, forge, authenticate, rescue, and reveal. It then proposes an institutional



Subset totals above; the full 12-event run has +3 trust and +21 influence/fear.

Figure 11. An audited two-round episode in Clockwork Conservatory. Six open actions form the central Marius–Valerius–Seraphina sequence; Lyra adds a seventh proposal about market effects. The seven-proposal subset has trust –29, affection –5, and influence/fear +17; the full run has different totals because it includes five other events.

loop: access to archives produces linguistic objects; translators and forgers can translate or forge them; diplomats exchange rights and debts; customs officers can inspect transfers; flood events threaten stored knowledge. The resulting world plan connects entities, conflicts, resource cycles, and open questions.

Planning establishes what must exist and why it matters. Later stages determine spatial layout, visual appearance, and resource quantities. Missing activity loops are refined before visual generation; for example, translation rights must have a source, a use, and a means of transfer.

Step 2: Semantic structure. Semantic generation expands the plan into rooms, roles, items, institutions, knowledge, and conflict hooks with stable identities. In this world, the graph contains eight rooms, including the Phoneme Exchange, Mnemonic Garden, Embassy Customs and Quarantine, and Ballast Hold; 25 named main characters; and 20 distinct items. Explicit relations connect each character to a home, each possession to an owner, and each action to relevant role, knowledge, location, or resource conditions.

The generation process also preserves asymmetry. Shen Mo is a Phonetic Smuggler and Lexical Restorationist based in the Cartographer’s Nook. His inventory includes a translation ear-cuff, archive key, forged exchange ledger, dive helmet, ritual ink, and a fragment of the Unspoken Treaty. His private knowledge includes phonetic decay patterns and ventilation shafts linking Customs to the Mnemonic Garden. Another character receives different property, authority, and information. This uneven allocation is what later makes persuasion, trade, secrecy, and coalition formation possible.

Step 3: Visual direction and cross-modal identity. The visual direction converts semantic identity into a shared canon: Art Nouveau diplomatic architecture, floating platforms, brass infrastructure, bioluminescent language media, water intrusion, and a melancholic ritual tone. It also emits per-entity appearance anchors. Room prompts inherit the canon but add function-specific materials; character prompts inherit world palette and wardrobe rules but retain role silhouettes; props receive icons whose shapes communicate their use. Shared style and entity-specific appearance jointly keep rooms, inhabitants, and objects recognizable as parts of the same world.

The visual direction guides each image independently while

maintaining common architecture, materials, palette, and role silhouettes. A visual inconsistency can therefore be refined locally while the semantic identity of the world remains stable.

Step 4: Evolving world state. This stage determines what can change after the images are produced. It creates wallets, inventories, property libraries, knowledge assets, directed relationships, task threads, room occupancy, and world-event status. It also defines invariants. Item quantities remain nonnegative; physical transfers conserve quantity; private knowledge has an owner and an acquisition event; a character occupies one reachable coordinate; relationship changes are directed. The generated repertoire and open-proposal forms are linked to the same world state.

The evolving state connects generated descriptions through observable causes and consequences. The treaty fragment in Shen Mo’s inventory can later be inspected, given away, stolen through an allowed rule, or used as an input to another action. When ownership changes, the next observation shows the new owner.

Step 5: Integrated world representation. The integrated representation connects semantic, visual, spatial, and evolving state. It supports cross-domain questions: Does every home room exist? Does every main character have a visual asset and a walkable placement? Is an item used by an action defined and obtainable? Does a private fact mention an existing institution? Does a room’s purpose agree with its objects and activities? These dependencies guide local refinement.

If a character image is inconsistent, roles, rooms, and items remain fixed. If an item relation is missing, only the connected content is regenerated. Figure 13 illustrates this process: two room images are refined while the rest of Tidal Embassy remains unchanged.

Step 6: Visual generation. FLUX generates the map source, individual room floors, characters, and local objects from the shared visual direction. Character images are arranged into directional movement atlases, and room images are placed within the planned regions and connected by traversable corridors. Figure 12 shows the composed map alongside three source floors, three character atlases, and an object.

Functional visibility connects visual quality to use: room imagery must cover its playable area, characters must remain recognizable while moving, and furniture must preserve navigable

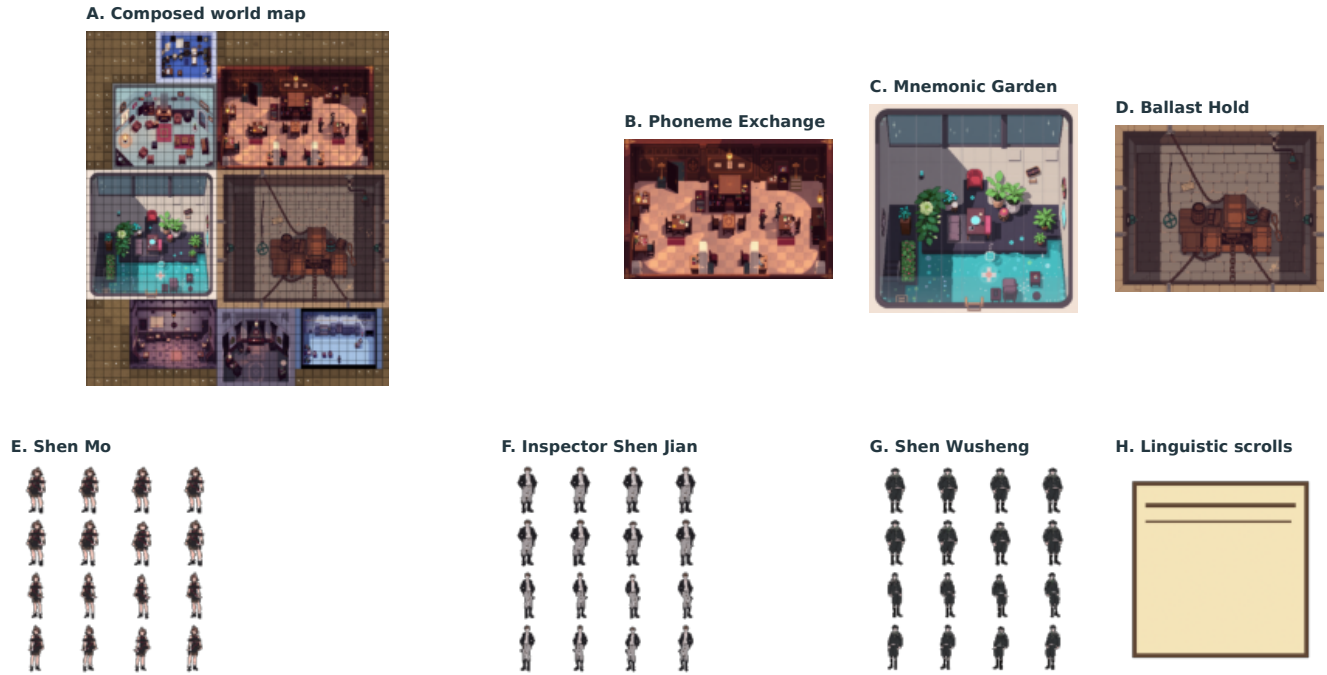


Figure 12. Generated visual elements from Tidal Embassy. The composed map contains eight generated floors (A–D show the map and three constituent rooms); E–G show the directional character atlases; H is a generated object. Shared visual motifs connect spatial, character, and object identity.

paths.

Step 7: Visual and spatial evaluation. The evaluation examines semantic completeness, visual identity, traversable geometry, foreground coverage, directional character consistency, readable room composition, and visibility during play. The Diplomat’s Lounge required three real generation attempts. Attempt 1 was classified as a poster card with an edge blank-background ratio of 0.56. Attempt 2 removed the card appearance but retained a blank-margin failure with a maximum blank-row ratio of 0.73. Only the third, edge-to-edge playable floor was accepted.

In Agora, visual quality combines coherent identity, readable composition, and fitness for navigation and interaction. Preserved refinement attempts make the progression from attractive illustration to usable world space observable.

Step 8: Persistent play. The generated world begins from a fixed initial state and evolves independently for each session. Humans and AI agents encounter the same rooms, inhabitants, objects, actions, and consequences. Separate world identities keep concurrent multiworld experiments isolated.

Figure 14 connects character generation to embodied play. A crop of the interaction capture shows the selected agent. A complementary world view shows neighboring agents among generated furniture and room surfaces, while the directional atlas records appearance across locomotion. The interactive trial covers movement, messaging, item use, trade, continued play, and a model-mediated reply.

The 25-character set uses one canonical visual identity per inhabitant. Front locomotion and side views preserve the same hair, clothing, body type, and palette, so an inhabitant remains recognizable while moving. Native 64×64 frames retain facial,

clothing, and limb detail in the stored atlases.

Character evaluation covers single-person composition, silhouette readability, palette contrast, directional identity, absence of text and detached effects, and visibility against the room surface. Contact-sheet review additionally examines the complete cast together, revealing cross-view identity changes and visual outliers that are difficult to judge one frame at a time.

Successive review passes identified characters with problematic props, enclosing effects, orientation, text, detached sparkles, or insufficient contrast. These counts reflect different review stages rather than disjoint character sets. Localized regeneration produced a final set of 25 coherent, readable character atlases, all visible during the interactive trial.

Step 9: Interaction-informed refinement. During play, agents and humans observe bounded state and choose either a world-specific action or an open proposal. The World Coordinator evaluates effects and enacts successful transitions, as detailed in Section 3. Interaction traces also provide feedback upstream. An action that is never selected may be poorly named or irrelevant; an unreachable character suggests placement repair; an important object that never enters an exchange suggests a missing action; frequent open proposals around one intention suggest a useful addition to the visible repertoire. Observed life inside the world can therefore guide the next round of world refinement.

Attempt 1: rejected



poster-card background
blank margin ratio: 0.56

Attempt 2: rejected



blank margin remains
max blank row: 0.73

Attempt 3: accepted



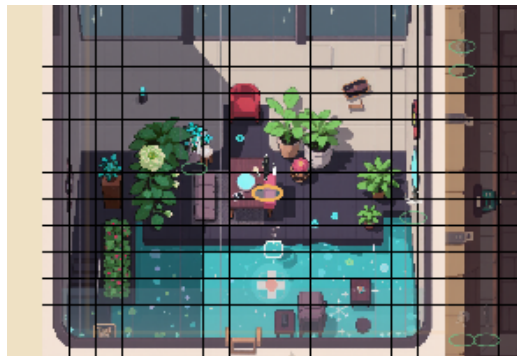
edge-to-edge playable floor
visual identity preserved

Figure 13. Localized visual refinement for the Diplomat's Lounge. The first image has a large poster-like margin; the second retains an extensive blank boundary; the third fills the playable floor while preserving the room's identity. Other rooms, characters, and world state remain unchanged.

A Selected agent



B Agents in the rendered world



C Directional atlas

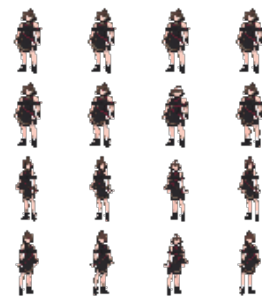


Figure 14. Character identity in Tidal Embassy. (A) A crop of the interaction capture shows the selected character, marked by an amber ring. (B) A complementary world view situates agents among generated room surfaces and furniture. (C) The directional atlas preserves clothing, palette, and silhouette across movement. The interactive trial includes movement, messaging, item use, and trade.